Status	Finished
Started	Thursday, 15 October 2026, 8:21 PM
Completed	Thursday, 15 October 2026, 8:24 PM
Duration	3 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Bubbling up the smallest element
c. Dividing the list into halves
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **2**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **4**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. sorted()
- c. arrange()
- d. order()

The correct answer is: sort()

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Merging two datasets
- b. Removing duplicates from a list
- c. Finding a specific element in a list
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **6**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By comparing elements sequentially
- c. By using the bubble-up method
- d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **7**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **8**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data in a single pass
- c. Sorting data sequentially
- d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Binary Search
- c. Linear Search
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Selection sort
- b. Merge sort
- c. Quick sort
- d. Heap sort

The correct answer is:
Merge sort

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Quick Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes data harder to manage
- c. Makes code execution slower
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It sorts data out of the box efficiently
- c. It requires external libraries
- d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It has no effect on the analysis process
- b. It allows for quicker identification of duplicates
- c. It makes the analysis slower
- d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **15**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It does not require recursion
- c. It is less efficient than bubble sort
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach